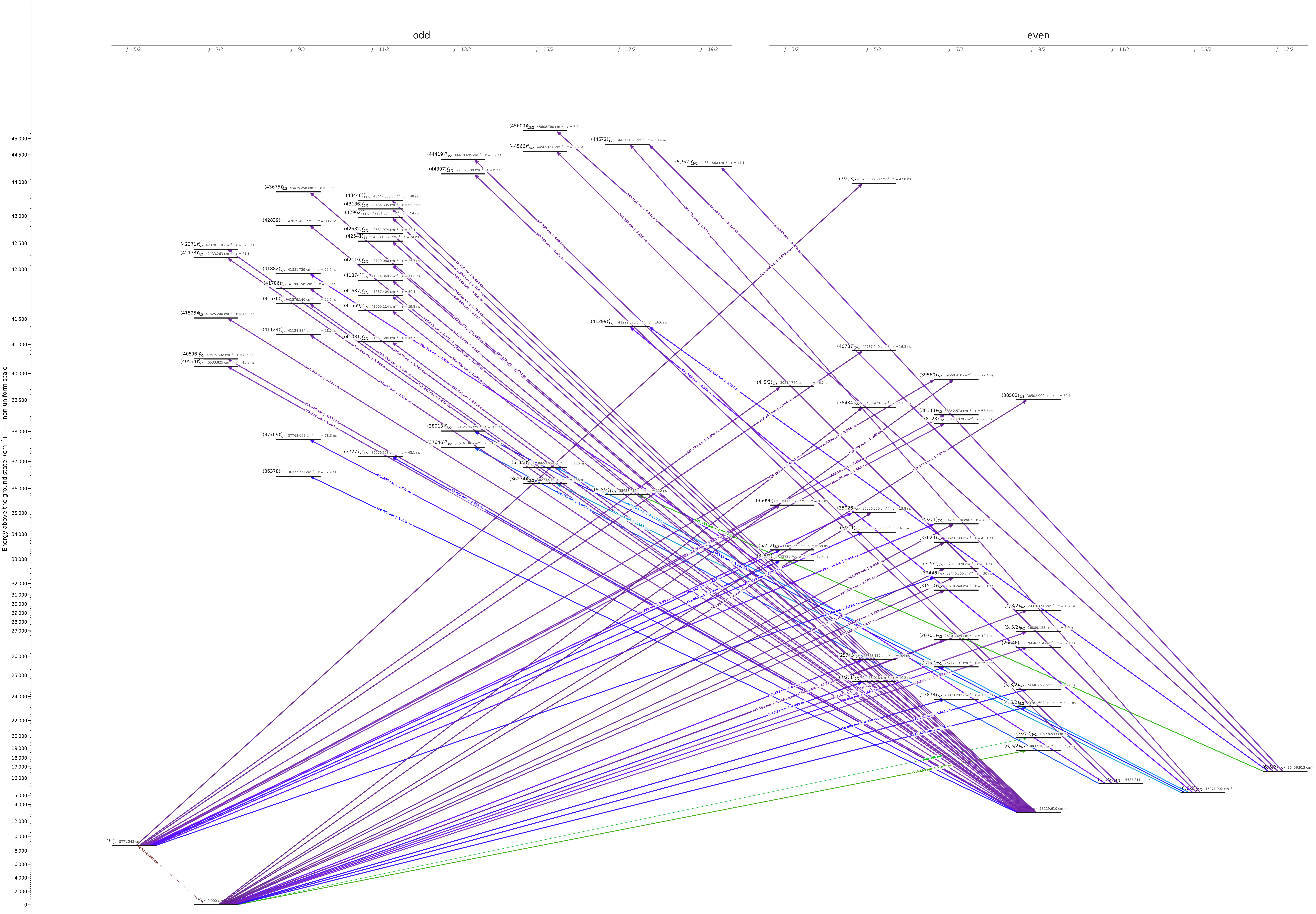
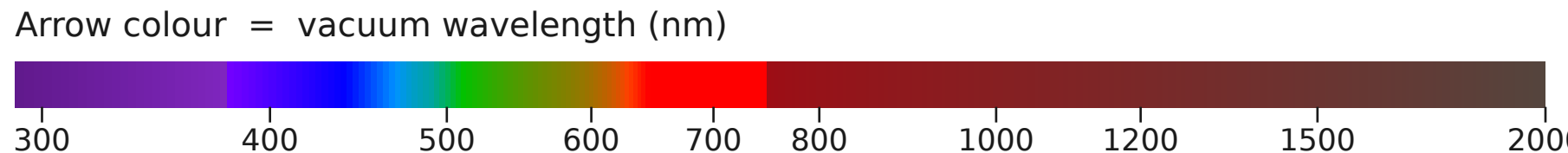


Thulium-169

¹⁶⁹Tm (Tm I), nuclear spin $I = 1/2$. Strongest classified transitions below 2 μm (NIST): 72 levels, 73 lines.

Tm⁺ ionisation limit 49880.57 cm⁻¹ = 6.18440 eV (200.479 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle U|er|U\rangle$ in ea0
Matrix elements are measured or NIST values: * = high-accuracy theory, $\approx$ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions								
Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\langle U er U\rangle$ (ea0)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
² F _{5/2} - (5, 3/2) _{o2}	410.6997	410.5838	729.9554	4.641	6.300e+07	10.14	100	first-stage
² F _{5/2} - (4, 5/2) _{o2}	420.4914	420.3730	712.9574	2.776	2.100e+07	3.872	86.3	alternative blue cycling line
² F _{5/2} - (7/2, 2) _{o2}	506.3665	506.2253	592.0464	-	-	-	-	third-stage deep narrow-line MOT
² F _{5/2} - (6, 5/2) _{o2}	530.8592	530.7116	564.7306	1.269	2.180e+06	0.3467	100	second-stage narrow-line
² F _{5/2} - ² F _{7/2}	1140.0896	1139.7776	262.955164	-	7.000e+00	1.421e-06	78.4	inner-shell M1 optical clock transition

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

¹⁶⁹ Tm hyperfine structure ($I = 1/2$)				
Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)	
² F _{5/2}	-374.137661(3)	-	F=3: +841.81	F=4: -654.74
² F _{7/2}	-704.648958	-	F=2: +1233.14	F=3: -880.81
(6, 5/2) _{o2}	-422.11(3)	-	F=4: +1160.81	F=5: -949.75
(4, 5/2) _{o2}	-317.32(16)	-	F=4: +872.63	F=5: -713.97
(23873) _{o2}	352.75(18)	-	F=3: -793.69	F=4: +617.31
(5, 3/2) _{o2}	-371.50(16)	-	F=4: +1021.62	F=5: -835.88
(7/2, 1) _{o2}	-656.5(4)	-	F=2: +1148.82	F=3: -820.58

Measured A, B constants; sources in the data file.